

DEVIKA YANAMALA

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Professional Summary

Data Science & AI professional with nearly 3 years of hands-on experience in data analytics using Python, SQL, and Power BI. Experienced in applying machine learning and AI-driven techniques to analyse data, build end-to-end data pipelines, and generate actionable insights in high-stakes environments. Skilled at translating complex datasets into clear, commercial stories and business recommendations for stakeholders.

Technical Expertise

- Programming Languages & Databases:** Python, SQL (PostgreSQL, Snowflake, MySQL, SQLite), R.
- Machine Learning & AI:** Supervised & Unsupervised Learning (Classification, Regression, Clustering), XGBoost, Scikit-Learn, SHAP (Explainable AI), NLP (spaCy, NLTK), LLM/Generative AI APIs (GPT-4, Gemini), Prompt Engineering.
- Data Engineering & Big Data:** ETL/ELT Pipelines, PySpark, Apache Kafka, Apache Spark, Data Warehousing, Data Validation & Auditing, API Integrations.
- Cloud & Infrastructure:** AWS (EC2, S3, Lambda), Git, Version Control, Docker (exposure).
- Business Intelligence & Analytics:** Power BI (Advanced DAX, Star Schema modeling, interactive dashboards), Excel (XLOOKUP, PivotTables, Power Query), Tableau, Matplotlib, Seaborn, Exploratory Data Analysis (EDA).
- Operations & Systems Analytics:** Capacity & Resource Planning, Production Scheduling, Inventory Tracking, Bottleneck Identification, Operational Efficiency Metrics.

Professional Experience

Global Industry Project

Newcastle University

June 2026 – July 2026

Industry Partner Project - Newcastle, UK

- Selected for a competitive, high-impact industry consulting program solving complex operational challenges for a UK industry partner.
- Collaborated in a team of data scientists to interrogate and analyze historical operational data, identifying critical process bottlenecks.
- Presented findings and recommendations to stakeholders, focusing on problem-solving and business impact.

Data Analyst – Product Analytics

Rooted Platform

04/2026 – Present

Independent Project - Remote

- Engineered a data analytics MVP using Python and SQL to ingest and analyze user behavior trends, user traffic patterns, and authentication logs, enabling data-driven product decisions.
- Designed and automated interactive Power BI/Streamlit reporting dashboards with advanced DAX measures, improving reporting efficiency and decision-making speed.
- Applied exploratory data analysis (EDA) and user segmentation to optimize pricing strategy, location-based store recommendations, and search features.

Data Analyst

Newcastle University

01/2026 – Present

Part-Time - Newcastle upon Tyne, UK

- Built robust Python ETL pipelines for automated data collection, cleaning, validation, and transformation, ensuring 100% data integrity for library services and occupancy analysis.
- Developed comprehensive KPI dashboards using Power BI to monitor real-time department occupancy, supporting strategic operational decisions.
- Interrogated large-scale occupancy datasets using SQL and Python to identify spatial usage trends, directly supporting capacity planning and optimizing department resource allocation.

Associate Data Scientist

BEPEC Solutions

05/2023 – 05/2025

Full-Time - Bangalore, India

- Developed, tuned, and deployed production-grade machine learning models (classification, regression) in Python, improving overall operational efficiency by 30% and saving 10+ hours per week.
- Analyzed large-scale structured datasets to identify commercial trends, performance inefficiencies, and logistics bottlenecks, translating complex results into clear business requirements.
- Partnered with product managers and data engineers in Agile sprints, leading client-facing meetings to gather analytics scope and deliver technical findings.

Data Scientist Intern

01/2023 – 05/2023

- Conducted extensive exploratory data analysis (EDA) and data preprocessing (imputation, encoding, scaling), contributing to a 15% improvement in forecasting accuracy by resolving data quality issues upstream.
- Created dynamic reporting dashboards in Power BI and Excel to track historical performance metrics, supporting resource planning and department scheduling.

Selected Projects & Initiatives

Commercial Opportunities Analytics Dashboard (Retail)

Personal Project | [GitHub](#)

- Built a Star Schema data model using Python (Pandas) to clean, transform, and optimize raw transactional sales data across 15 European markets.
- Developed an executive Power BI dashboard with 20+ advanced DAX measures (time intelligence, segment grouping) to track key retail metrics.
- Analyzed pricing and customer segments to identify discount optimization opportunities, demonstrating over £150K+ in annual profit recovery.

Pharmaceutical Marketing Strategy & Attribution Analysis

Personal Project | [GitHub](#)

- Built a regression-based marketing attribution model in Python to estimate the commercial impact and ROI of three distinct campaign strategies.
- Applied Interrupted Time Series (ITS) analysis and statistical testing to evaluate competitor market entry and predict sales deceleration.
- Designed an interactive Power BI Strategy Simulator using DAX what-if parameters, identifying high-impact channels and driving \$1.3B+ in attributed revenue.

Next Best Action & Churn Prediction Model (Banking)

Personal Project | [GitHub](#)

- Developed an end-to-end machine learning pipeline using Python (Scikit-Learn, Imbalanced-Learn) to predict customer churn, resolving class imbalance with SMOTE.
- Engineered a Next Best Action (NBA) recommendation algorithm that maps customer transaction patterns to tailored retention products, reducing manual review overhead.
- Applied SHAP (SHapley Additive exPlanations) to ensure model transparency and explainability, aligning with UK banking regulatory standards.

Education

Master of Science (MSc), Data Science & Artificial Intelligence

Expected 09/2026

Newcastle University

Newcastle upon Tyne, UK

- **UK Coursework:** Data Visualisation, Machine Learning, Statistical Foundations of Data Science, Deep Learning, Image Processing, Generative AI for Business, Advanced AI, Group Project.
- **Dissertation:** Towards the Modelling of Traffic Flows from Open-Source Data – An Assessment of Automated OSM Topological Verification for Machine Learning-Based AADT Prediction. Developing a python pipeline (OSMnx, NetworkX) to resolve topological errors and engineer graph-theoretic centrality features to improve Random Forest and XGBoost traffic prediction models.

Bachelor of Technology

2019 – 2023

IIIT, RGUKT

CGPA: 8.7/10 - RK Valley, India

Professional Certifications

UK Work Culture Programme (Awarded Participant)

Awarded 2026

Newcastle University

Professional Development Programme

- Selected for a competitive programme focused on UK industry standards and corporate communication, presenting business solutions to stakeholders.

Additional Experience & Recognition

Global Ambassador

05/2023 – Present

WomenTech Network

Volunteering Leadership Role

- Active participant in global initiatives promoting technical excellence, leadership, and diversity in AI and Data Science.

Volunteer Module Representative

05/2026

Newcastle University

Elected Student Leadership Role

- Elected to represent the MSc cohort, dedicating time to collaborate directly with faculty to optimize curriculum delivery and the student experience.